

Batch : DV Fri / C-3 Roll No.: 16010123325

Experiment No. 1

Grade: AA / AB / BB / BC / CC / CD / DD

TITLE: To clean real-world data using Excel

Objective:

To clean and preprocess a raw dataset using Microsoft Excel by handling the missing values, inconsistencies, invalid entries, and duplicate records using built-in functions.

Expected OUTCOME of Experiment: CO 1

Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats

Books/ Journals/ Websites referred:

1. Data Analysis with Microsoft Excel, Wayne L. Winston, Microsoft Press.
2. Excel 2019 Bible, John Walkenbach, Wiley.
3. Practical Statistics for Data Scientists, Peter Bruce and Andrew Bruce, O'Reilly Media

Background Theory:

Data cleaning is a crucial step in the data preprocessing stage of data analysis and machine learning. Real-world datasets are often incomplete, inconsistent, and contain errors due to manual data entry, system faults, or data integration from multiple sources. Such issues reduce data quality and can lead to incorrect analysis, misleading visualizations, and inaccurate model predictions.

Microsoft Excel is widely used for data cleaning due to its simplicity, availability, and powerful built-in functions. Excel allows users to identify and correct common data quality issues such as missing values, duplicate records, inconsistent formatting, invalid data entries, and unwanted characters without requiring programming knowledge. One

of the most common data cleaning tasks is the removal of extra spaces and formatting inconsistencies in textual data. Functions such as `TRIM`, `UPPER`, `LOWER`, and `PROPER` are used to standardize text fields like names, cities, and categorical values. This ensures uniform representation of data and prevents logical errors during comparison and analysis.

Handling missing values is another important aspect of data cleaning. Blank or null values can disrupt calculations and analytical processes. Excel functions such as `IF` and `ISBLANK` are used to detect missing values and replace them with suitable default values, such as averages or domain-specific constants. This process is known as data imputation.

Data validation is essential to ensure that numerical values fall within acceptable ranges. Logical functions like `IF`, `AND`, and `OR` are used to detect invalid values such as negative marks or values exceeding defined limits. Similarly, basic validation of textual fields like email addresses can be performed using functions such as `SEARCH` and `ISERROR`. Duplicate records are a common issue in datasets collected from multiple sources or through repeated entries. Excel provides functions like `COUNTIF` to identify duplicate values and built-in tools such as ****Remove Duplicates**** to eliminate redundant records, thereby improving data integrity.

Data cleaning in Excel involves a combination of text processing, logical decision-making, validation, and duplication handling. Performing proper data cleaning ensures data accuracy, consistency, and reliability, making the dataset suitable for further analysis, visualization, and machine learning applications.

Laboratory Work:

Students will work on a raw excel sheet, apply data cleaning functions and create a new clean excel sheet.

Step 1- Download the excel file from this link- [Excel Dataset Folder](#)

Step 2- Write the issues/ problems with the raw excel file

Raw Excel File Screenshot

Student_ID	Name	Age	Gender	Email	Marks	City
101	Rahul	20	M	rahul@gmail.com	85	Mumbai
102	priya	19	F	priya@@gmail.com	90	mumbai
103	Ankit		M	ankit@gmail	78	Pune
104	Sneha Diwa	21	Female	sneha@gmail.com	-5	Delhi
105	NULL	22	F		88	Chennai
106	Ramesh123	20	M	ramesh@gmail.com	105	Bangalore
107	Kavita	19	f	kavita@gmail.com	92	bangalore
107	Kavita	19	f	kavita@gmail.com	92	bangalore

Expected clean Excel sheet after applying all Functions (end of this lab):

Student_ID	Name	Age	Gender	Email	Marks	City
101	Rahul	20	Male	rahul@gmail.com	85	Mumbai
102	Priya	19	Female	Invalid	90	Mumbai
103	Ankit	20	Male	Invalid	78	Pune
104	Sneha	21	Female	sneha@gmail.com	0	Delhi
105	Unknown	22	Female	Missing	88	Chennai
106	Ramesh	20	Male	ramesh@gmail.com	100	Bangalore
107	Kavita	19	Female	kavita@gmail.com	92	Bangalore

Step 3- Use Functions on raw excel sheet

3.1 TRIM()

Purpose: Removes extra spaces from text.

Syntax:

=TRIM(text)

Steps:

1. Select a new column.
2. Enter =TRIM(cell_reference).
3. Press Enter and drag down.

Example:

=TRIM(B2)

3.2 PROPER()

Purpose: Converts text to proper case.

Syntax:

=PROPER(text)

Steps:

1. Apply after TRIM().
2. Enter =PROPER(cell_reference).

Example:

=PROPER(TRIM(B2))

3.3 IF() and ISBLANK()

Purpose: Handles missing values.

Syntax:

=IF(ISBLANK(cell), value_if_blank, cell)

Here, **ISBLANK(C2)** checks if cell C2 has no value, and returns:

TRUE → if C2 is empty

FALSE → if C2 has some value

And, **IF(condition, value_if_true, value_if_false)** is the decision-making function.

Hence,

IF(
ISBLANK(C2), ← condition
20, ← if TRUE
C2 ← if FALSE
)

Interpretation: If C2 cell is blank, insert 20 value in it, else keep the original value intact.

Steps:

1. Identify columns with missing values.
2. Replace blanks with default values.

Example:

=IF(ISBLANK(C2), 20, C2)

3.4 SUBSTITUTE()

Purpose: Removes unwanted characters from a single cell

Syntax:

=SUBSTITUTE(cell, old_text, new_text)

Example:

=SUBSTITUTE(B2,"Ramesh123","Ramesh")

3.5 IF() with OR()

Purpose: Validates numeric ranges in a cell and this function can be dragged to the other cells in the column

Syntax:

=IF(OR(condition1, condition2), value_if_true, value_if_false)

Example:

=IF(OR(F2<0, F2>100), 0, F2)

Interpretation:

If the value in cell F2 is less than 0 or greater than 100 (i.e. TRUE condition), then replace the value with 0, else keep it as it is.

3.6 SEARCH() and ISERROR()

Purpose: Validates email format

Syntax:

=IF (ISERROR(SEARCH("@",E2)), "Invalid", E2)

This formula checks whether an email contains both @ and . and marks it invalid if either is missing.

=IF (OR(ISERROR(SEARCH ("@",E2)), ISERROR (SEARCH(".",E2))), "Invalid", E2)

Interpretation: Let's break down the formula for clear understanding

```
=IF(
  OR(
    ISERROR(SEARCH("@", E2)),
    ISERROR(SEARCH(".", E2))
  ),
  "Invalid",
  E2
)
```

SEARCH("@", E2) - Looks for the @ symbol inside cell E2.

If found → returns the position number

If NOT found → returns an error

Example:

SEARCH("@", "rahul@gmail.com") → 6

SEARCH("@", "rahulgmail.com") → error

ISERROR(SEARCH("@", E2))- Checks whether SEARCH("@",E2) produced an error and returns:

FALSE → @ exists

TRUE → @ missing

OR (ISERROR(SEARCH("@", E2)), ISERROR(SEARCH(".",E2)))- Returns TRUE if any one of these is true:

@ is missing *OR*

. is missing

Final IF() decision- IF(condition, value_if_true, value_if_false) *implies* If either @ or . is missing → "Invalid",

Else → return the original email

Let's assume input is **ankit@gmail**

Check	Result
@ exists?	YES
. exists?	NO
OR condition	TRUE
Output	Invalid

3.7 COUNTIF()

Purpose: counts the number of cells that meet a specified condition.

Syntax:

=COUNTIF(range, criteria)

Example:

=COUNTIF(A:A,A2)

Interpretation: Counts the number of cells that have value equal to value in A2 cell.

Try different versions of COUNTIF

- Count how many times the value in A2 appears in a column
=COUNTIF(A:A, A2)
- Count non-empty cells
=COUNTIF(A:A, "<>")
- Count blank cells
=COUNTIF(A:A, "")
- Check if a value exists or not
=IF(COUNTIF(A:A, A2)>0, "Exists", "Does not exist")

3.8 CLEAN()

Purpose: Data copied from websites or PDFs may contain hidden characters that TRIM cannot remove. This helps remove Non-Printable Characters. It removes line breaks and non-visible characters.

Syntax:

=CLEAN(text)

Example

=CLEAN(B2)

If B2 contains-

tom

orrow (note there is line break)

It will be converted to **tomorrow**

3.9 LEN()

Purpose: It returns the number of characters in a text string. It helps validate data length, especially for: Phone numbers, IDs, Short/empty names

Syntax:

=LEN(text)

Example (if length is less than 3, returns "Invalid Name", else keeps value intact)

=IF(LEN(B2)<3, "Invalid Name", B2)

3.10 IFERROR()

Purpose: It handles errors cleanly. It replaces formula errors with a user-defined value. When formulas like SEARCH() or division operations fail, Excel shows errors (#VALUE!, #DIV/0!). IFERROR() replaces these errors with meaningful values.

Syntax:

=IFERROR(value, value_if_error)

Example (Email validation)

=IFERROR(SEARCH("@",E2),"Invalid")

Dataset: healthcare_data

Excel Book : <https://docs.google.com/spreadsheets/d/17sf--QrElqlfL-dXaiBugFvEvgcsiW9b/edit?usp=sharing&ouid=107889401080401174487&rtpof=true&sd=true>

1) Patient Name

Clean & Format

AutoSave healthcare_messy_data

Home Insert Draw Page Layout Formulas Data Review View

Calibri (Body) 12 A+ A- B I U Merge & Centre Wrap Text General Conditional Formatting Format as Table Cell Styles Insert Delete Format Comments Sort & Filter Find & Select Analyse Data

Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Save As...

	A	B	C	D	E	F	G	H	I	J	K
	Patient Name	Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number	
1	David Lee	25	Other	Heart Disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555	
2	Emily Davis	nan	Male	Diabetes	NONE	April 5, 2018	120/80	200	nan	nan	
3	Laura Martinez	35	Other	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan	
4	Michael Wilson	nan	Male	Diabetes	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555	
5	David Lee	nan	Female	Asthma	NONE	20/02/20	110/70	180	nan	nan	
6	Mary Clark	nan	Male	Hypertension	METFORMIN	03-25-2019	140/90	180	name@hospital.org	nan	
7	Robert Brown	nan	Male	Hypertension	LISINAPRIL	20/02/20	120/80	NaN	name@hospital.org	nan	
8	David Lee	60	Other	Heart Disease	NONE	20/02/20	140/90	180	name@hospital.org	nan	
9	Laura Martinez	forty	Female	Asthma	ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555	
10	Michael Wilson	25	Male	Diabetes	ATORVASTATIN	April 5, 2018	120/80	200	nan	nan	
11	Emily Davis	nan	Male	Diabetes	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org	nan	
12	Laura Martinez	forty	Male	Diabetes	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890	
13	Robert Brown	25	Female	Asthma	ALBUTEROL	2019.12.01	130/85	NaN	name@hospital.org	nan	
14	Sarah Johnson	70	Female	Hypertension	ALBUTEROL	2019.12.01	130/85	220	nan	nan	
15	Michael Wilson	35	Female	Heart Disease	ATORVASTATIN	03-25-2019	110/70	180	nan	123-456-7890	
16	Jane Smith	60	Female	Asthma	ALBUTEROL	20/02/20	120/80	200	nan	nan	
17	Laura Martinez	70	Female	Asthma	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321	
18	Sarah Johnson	25	Other	Diabetes	NONE	01/15/2020	120/80	NaN	name@hospital.org	098-765-4321	
19	Jane Smith	70	Male	Heart Disease	NONE	03-25-2019	110/70	160	contact@domain.com	nan	
20	Michael Wilson	35	Male	Hypertension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan	
21	John Doe	25	Male	Asthma	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321	
22	Mary Clark	70	Male	Asthma	NONE	2019.12.01	130/85	220	nan	555-555-5555	
23	Sarah Johnson	60	Male	Diabetes	METFORMIN	April 5, 2018	120/80	160	nan	555-555-5555	
24	James Taylor	60	Male	None	ATORVASTATIN	20/02/20	NaN	220	nan	123-456-7890	
25	John Doe	forty	Male	Hypertension	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com	nan	
26	Mary Clark	70	Male	Heart Disease	METFORMIN	20/02/20	130/85	160	name@hospital.org	nan	
27	Robert Brown	nan	Other	Asthma	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321	
28	David Lee	70	Male	Asthma	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan	
29	Emily Davis	25	Other	Hypertension	NONE	April 5, 2018	130/85	160	name@hospital.org	nan	
30	James Taylor	25	Female	None	NONE	01/15/2020	140/90	200	nan	098-765-4321	
31	Robert Brown	70	Male	Asthma	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321	
32	Michael Wilson	35	Female	Hypertension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org	nan	
33	Robert Brown	60	Female	Diabetes	LISINAPRIL	03-25-2019	110/70	160	nan	098-765-4321	
34	David Lee	25	Female	None	LISINAPRIL	20/02/20	130/85	NaN	name@hospital.org	123-456-7890	
35	Michael Wilson	60	Other	Heart Disease	ATORVASTATIN	2019.12.01	140/90	NaN	nan	123-456-7890	
36	James Taylor	25	Female	Asthma	LISINAPRIL	01/15/2020	110/70	200	contact@domain.com	nan	
37	David Lee	forty	Female	Hypertension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan	

Ready Accessibility: Unavailable Count: 1001 100%

Process: TRIM() -> PROPER()

L2 =PROPER(TRIM(A2))

	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Condition	Medication	Visit Date	Blood Pressu	Cholesterol	Email	Phone Number		David Lee						
2	Heart Disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555		Emily Davis						
3	Diabetes	NONE	April 5, 2018	120/80	200	nan			Laura Martinez						
4	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan		Michael Wilson						
5	Diabetes	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555								

2) Age

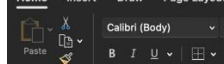
Change 'nan' values to blank , “forty” string converted to 40

	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	on	Medication	Visit Date	Blood Pressu	Cholesterol	Email	Phone Number									
2	disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555		=IF(B2="", "",							
3	is	NONE	April 5, 2018	120/80	200	nan	nan		IFERROR(VALUE(B2),							
4	h	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan		IF (LOWER(B2)~"forty",40,""))							
5	is	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555		=IF(logical, (value_if_true), (value_if_false))							
6	h	NONE	20/02/20	110/70	180	nan	nan									
7	ension	METFORMIN	03-25-2019	140/90	180	nan	nan									
8	ension	LISINAPRIL	20/02/20	120/80	NaN	name@hospital.org	nan									
9		NONE	20/02/20	NaN	NaN	name@hospital.org	nan									
10	disease	NONE	01/15/2020	140/90	180	name@hospital.org	nan									
11		ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555									
12		ATORVASTATIN	April 5, 2018	130/80	200	nan	nan									
13		ATORVASTATIN	20/02/20	130/85	200	name@hospital.org	nan									
14	is	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890									
15		ALBUTEROL	03-25-2019	130/85	NaN	name@hospital.org	nan									
16	ension	ALBUTEROL	2019.12.01	130/85	220	nan	nan									
17	disease	ATORVASTATIN	03-25-2019	110/70	180	nan	123-456-7890									
18		ALBUTEROL	20/02/20	120/80	200											
19		METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321									
20	is	NONE	01/15/2020	120/80	NaN	name@hospital.org	098-765-4321									
21	disease	NONE	03-25-2019	110/70	160	contact@domain.com	nan									
22	ension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan									
23		LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321									
24	is	NONE	2019.12.01	130/85	220	nan	555-555-5555									
25		METFORMIN	April 5, 2018	120/80	220	nan	555-555-5555									
26		ATORVASTATIN	20/02/20	NaN	200	nan	123-456-7890									
27		ATORVASTATIN	20/02/20	120/80	200	contact@domain.com										
28	disease	METFORMIN	20/02/20	130/85	160	name@hospital.org										
29		ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321									
30		ALBUTEROL	April 5, 2018	110/70	220	name@domain.com	nan									
31	ension	NONE	April 5, 2018	130/85	160	name@hospital.org										
32		NONE	01/15/2020	140/90	200	nan	098-765-4321									
33		METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321									
34	ension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org										
35	is	LISINAPRIL	03-25-2019	110/70	160		098-765-4321									
36		LISINAPRIL	20/02/20	130/85	NaN	name@hospital.org	123-456-7890									
37	disease	ATORVASTATIN	2019.12.01	140/90	nan	contact@domain.com	123-456-7890									
38		LISINAPRIL	110/70	200	contact@domain.com											
39	ension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan									

Process: IF() , IFERROR()

AutoSave  healthcare_messy_data Search (Cmd + Ctrl + U)

Home Insert Draw Page Layout Formulas Data Review View Comments Share

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Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Save As...

L2                                                           

3) Medication

Format

Possible Data Loss Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Save As...

	A	B	C	D	E	F	G	H	I	J
	Patient Name	Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number
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2	Emily Davis		Male	Diabetes	NONE	April 5, 2018	120/80	200	nan	nan
3	Laura Martinez	35	Other	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan
4	Michael Wilson		Male	Diabetes	ALBUTEROL	01/15/2020	110/70	180	name@hospital.org	555-555-5555
5	David Lee		Female	Asthma	NONE	20/02/20	110/70	180	nan	nan
6	Mary Clark		Male	Hypertension	METFORMIN	03-25-2019	140/90	180	name@hospital.org	nan
7	Robert Brown		Male	Hypertension	LISINAPRIL	20/02/20	120/80		name@hospital.org	nan
8	David Lee	60	Other	None	NONE	20/02/20			name@hospital.org	nan
9	Laura Martinez	60	Other	Heart Disease	NONE	01/15/2020	140/90	180	name@hospital.org	nan
10	Michael Wilson	40	Female	None	ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555
11	Emily Davis	25	Male	Asthma	ATORVASTATIN	April 5, 2018	120/80	200	nan	nan
12	Laura Martinez		Male	None	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org	nan
13	Laura Martinez	40	Male	Diabetes	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890
14	Robert Brown	25	Female	Asthma	ALBUTEROL	03-25-2019	130/85		name@hospital.org	nan
15	Sarah Johnson	70	Female	Hypertension	ALBUTEROL	2019.12.01	130/85	220		nan
16	Michael Wilson	35	Female	Heart Disease	ATORVASTATIN	03-25-2019	110/70	180		123-456-7890
17	Jane Smith	60	Female	Asthma	ALBUTEROL	20/02/20	120/80	200		nan
18	Laura Martinez	70	Female	Asthma	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321
19	Sarah Johnson	25	Other	Diabetes	NONE	01/15/2020	120/80	200	name@hospital.org	098-765-4321
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21	Michael Wilson	35	Male	Hypertension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan
22	John Doe	25	Male	Asthma	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321
23	Mary Clark	70	Male	Asthma	NONE	2019.12.01	130/85	220	nan	555-555-5555
24	Sarah Johnson	60	Male	Diabetes	METFORMIN	April 5, 2018	120/80	160	nan	555-555-5555
25	James Taylor		Male	None	ATORVASTATIN	20/02/20		200	nan	123-456-7890
26	John Doe	40	Male	Hypertension	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com	
27	Mary Clark	70	Male	Heart Disease	METFORMIN	20/02/20	130/85	160	name@hospital.org	
28	Robert Brown		Other	Asthma	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321
29	David Lee	70	Male	Asthma	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan
30	Emily Davis	25	Other	Hypertension	NONE	April 5, 2018	130/85	160	name@hospital.org	
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32	Robert Brown	70	Male	Asthma	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321
33	Michael Wilson	35	Female	Hypertension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org	
34	Robert Brown	60	Female	Diabetes	LISINAPRIL	03-25-2019	110/70	160		098-765-4321
35	David Lee	25	Female	None	LISINAPRIL	20/02/20	130/85		name@hospital.org	123-456-7890
36	Michael Wilson	60	Other	Heart Disease	ATORVASTATIN	2019.12.01	140/90		contact@domain.com	123-456-7890
37	James Taylor	40	Female	Asthma	LISINAPRIL	01/15/2020		200	contact@domain.com	
38	David Lee	25	Female	Hypertension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan

Ready Accessibility: Unavailable Count: 1001 100%

Process: TRIM()+PROPER()

	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number		=PROPER(TRIM(E2))						
1	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555		Metformin						
2	NONE	April 5, 2018	120/80	200	nan			None						
3	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan		Metformin						
4	ALBUTEROL	01/15/2020	110/70	180	name@hospital.org	555-555-5555		Albuterol						
5	NONE	20/02/20	110/70	180				None						
6	METFORMIN	03-25-2019	140/90	180	nan	nan		Metformin						
7	LISINAPRIL	20/02/20	120/80		name@hospital.org	nan		Lisinopril						
8	NONE	20/02/20	140/90	180	name@hospital.org	nan		None						
9	NONE	01/15/2020	140/90	200	patient@example.com	555-555-5555		None						
10	ALBUTEROL	20/02/20	140/90	220	nan			Albuterol						
11	ATORVASTATIN	April 5, 2018	120/80	200	nan			Atorvastatin						
12	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org			Atorvastatin						
13	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890		Albuterol						
14	ALBUTEROL	03-25-2019	130/85		name@hospital.org	nan		Albuterol						
15	ALBUTEROL	2019.12.01	130/85	220				Albuterol						
16	ATORVASTATIN	03-25-2019	110/70	180	NONE	123-456-7890		Atorvastatin						
17	ALBUTEROL	20/02/20	120/80	200				Albuterol						
18	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321		Metformin						
19	NONE	01/15/2020	120/80		name@hospital.org	098-765-4321		None						
20	NONE	03-25-2019	110/70	160	contact@domain.com			None						
21	LISINAPRIL	April 5, 2018	120/80	180	nan	nan		Lisinopril						
22	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321		Lisinopril						
23	NONE	2019.12.01	130/85	220	nan	555-555-5555		None						
24	METFORMIN	April 5, 2018	120/80	160	nan	555-555-5555		Metformin						
25	ATORVASTATIN	20/02/20	120/80	220	nan	123-456-7890		Atorvastatin						
26	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com			Atorvastatin						
27	METFORMIN	20/02/20	130/85	160	name@hospital.org			Metformin						
28	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321		Atorvastatin						
29	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan		Albuterol						
30	NONE	April 5, 2018	130/85	160	name@hospital.org			None						
31	NONE	01/15/2020	140/90	200	nan	098-765-4321		None						
32	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321		Metformin						
33	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org			Atorvastatin						
34	LISINAPRIL	03-25-2019	110/70	160	name@hospital.org	098-765-4321		Lisinopril						
35	LISINAPRIL	20/02/20	130/85		name@hospital.org	123-456-7890		Lisinopril						
36	ATORVASTATIN	2019.12.01	140/90			123-456-7890		Atorvastatin						
37	LISINAPRIL	01/15/2020	110/70	200	contact@domain.com			Lisinopril						
38	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan		Metformin						

Edit Accessibility: Unavailable 100%

4) Blood Pressure

Remove NaN

G9										
	A	B	C	D	E	F	G	H	I	J
1	Patient Name	Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number
2	David Lee	25	Other	Heart Disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555
3	Emily Davis	35	Male	Diabetes	NONE	April 5, 2018	120/80	200	nan	nan
4	Laura Martinez	35	Other	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan
5	Michael Wilson	Male	Diabetes	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555	nan
6	David Lee	Female	Asthma	NONE	03-25-2019	20/02/20	110/70	180	nan	nan
7	Mary Clark	Male	Hypertension	METFORMIN	03-25-2019	20/02/20	140/90	NaN	name@hospital.org	nan
8	Robert Brown	Male	Hypertension	LISINAPRIL	03-25-2019	20/02/20	120/80	NaN	name@hospital.org	nan
9	David Lee	60	Other	None	NONE	20/02/20	NaN	NaN	name@hospital.org	nan
10	Laura Martinez	60	Other	Heart Disease	NONE	01/15/2020	140/90	nan	180	name@hospital.org
11	Michael Wilson	40	Female	None	ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555
12	Emily Davis	25	Male	Asthma	ATORVASTATIN	April 5, 2018	120/80	200	nan	nan
13	Laura Martinez	Male	None	Asthma	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org	nan
14	Laura Martinez	40	Male	Diabetes	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890
15	Robert Brown	25	Female	Asthma	ALBUTEROL	03-25-2019	130/85	NaN	name@hospital.org	nan
16	Sarah Johnson	70	Female	Hypertension	ALBUTEROL	2019.12.01	130/85	220	nan	nan
17	Michael Wilson	35	Female	Heart Disease	ATORVASTATIN	03-25-2019	110/70	180	nan	123-456-7890
18	Jane Smith	60	Female	Asthma	ALBUTEROL	20/02/20	120/80	200	nan	nan
19	Laura Martinez	70	Female	Asthma	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321
20	Sarah Johnson	25	Other	Diabetes	NONE	01/15/2020	120/80	NaN	name@hospital.org	098-765-4321
21	Jane Smith	70	Male	Heart Disease	NONE	03-25-2019	110/70	160	contact@domain.com	nan
22	Michael Wilson	35	Male	Hypertension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan
23	John Doe	25	Male	Asthma	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321
24	Mary Clark	70	Male	Asthma	NONE	2019.12.01	130/85	220	nan	555-555-5555
25	Sarah Johnson	60	Male	Diabetes	METFORMIN	April 5, 2018	120/80	160	nan	555-555-5555
26	James Taylor	60	Male	None	ATORVASTATIN	20/02/20	NaN	220	nan	123-456-7890
27	John Doe	40	Male	Hypertension	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com	nan
28	Mary Clark	70	Male	Heart Disease	METFORMIN	20/02/20	130/85	160	name@hospital.org	nan
29	Robert Brown	Other	Asthma	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321	nan
30	David Lee	70	Male	Asthma	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan
31	Emily Davis	25	Other	Hypertension	NONE	April 5, 2018	130/85	160	name@hospital.org	nan
32	James Taylor	25	Female	None	NONE	April 5, 2018	130/85	200	name@hospital.org	098-765-4321
33	Robert Brown	70	Male	Asthma	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321
34	Michael Wilson	35	Female	Hypertension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org	nan
35	Robert Brown	60	Female	Diabetes	LISINAPRIL	03-25-2019	110/70	160	nan	098-765-4321
36	David Lee	25	Female	None	LISINAPRIL	20/02/20	130/85	NaN	name@hospital.org	123-456-7890
37	Michael Wilson	60	Other	Heart Disease	ATORVASTATIN	2019.12.01	140/90	NaN	nan	123-456-7890
38	James Taylor	25	Female	Asthma	LISINAPRIL	01/15/2020	110/70	200	contact@domain.com	nan
39	David Lee	40	Female	Hypertension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan

Process: IF()

TRIM										
	C	D	E	F	G	H	I	J	K	L
1	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number			
2	Heart Disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555			
3	Diabetes	NONE	April 5, 2018	120/80	200	nan	nan			
4	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan			
5	Diabetes	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555			
6	Asthma	NONE	20/02/20	110/70	180	nan	nan			
7	Hypertension	METFORMIN	03-25-2019	140/90	NaN	name@hospital.org	nan			
8	Hypertension	LISINAPRIL	03-25-2019	120/80	NaN	name@hospital.org	nan			
9	None	NONE	20/02/20	NaN	NaN	name@hospital.org	nan			
10	Heart Disease	NONE	01/15/2020	140/90	180	nan	nan			
11	None	ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555			
12	Asthma	ATORVASTATIN	April 5, 2018	120/80	200	nan	nan			
13	None	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org	nan			
14	Diabetes	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890			
15	Asthma	ALBUTEROL	03-25-2019	130/85	NaN	name@hospital.org	nan			
16	Hypertension	ALBUTEROL	2019.12.01	130/85	220	nan	nan			
17	Heart Disease	ATORVASTATIN	03-25-2019	110/70	180	nan	123-456-7890			
18	Asthma	ALBUTEROL	20/02/20	120/80	200	nan	nan			
19	Asthma	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321			
20	Diabetes	NONE	01/15/2020	120/80	NaN	name@hospital.org	098-765-4321			
21	Heart Disease	NONE	03-25-2019	110/70	160	contact@domain.com	nan			
22	Hypertension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan			
23	Asthma	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321			
24	Asthma	NONE	2019.12.01	130/85	220	nan	555-555-5555			
25	Diabetes	METFORMIN	April 5, 2018	120/80	160	nan	555-555-5555			
26	None	ATORVASTATIN	20/02/20	NaN	220	nan	123-456-7890			
27	Hypertension	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com	nan			
28	Heart Disease	METFORMIN	20/02/20	130/85	160	name@hospital.org	nan			
29	Asthma	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321			
30	Asthma	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan			
31	Hypertension	NONE	April 5, 2018	130/85	160	name@hospital.org	nan			
32	None	NONE	01/15/2020	140/90	200	nan	098-765-4321			
33	Asthma	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321			
34	Hypertension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org	nan			
35	Diabetes	LISINAPRIL	03-25-2019	110/70	160	nan	098-765-4321			
36	None	LISINAPRIL	20/02/20	130/85	NaN	name@hospital.org	123-456-7890			
37	Heart Disease	ATORVASTATIN	2019.12.01	140/90	NaN	nan	123-456-7890			
38	Asthma	LISINAPRIL	01/15/2020	110/70	200	contact@domain.com	nan			
39	Hypertension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan			

5) Cholesterol

Remove NaN values

H1	B	C	D	E	F	G	H	I	J	K	L
	Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number		
1	25	Other	Heart Disease	METFORMIN	01/15/2020	140/90	200	name@hospital.org	555-555-5555		
2		Male	Diabetes	NONE	April 5, 2018	120/80	200	nan	nan		
3	35	Other	Asthma	METFORMIN	2019.12.01	110/70	160	contact@domain.com	nan		
4		Female	Diabetes	ALBUTEROL	01/15/2020	110/70	NaN	name@hospital.org	555-555-5555		
5		Male	Asthma	NONE	20/02/20	110/70	180				
6		Male	Hypertension	METFORMIN	03-25-2019	140/90	180	nan	nan		
7		Male	Hypertension	LISINAPRIL	20/02/20	120/80	NaN	name@hospital.org	nan		
8	60	Other	None	NONE	20/02/20	140/90	NaN	name@hospital.org	nan		
9	60	Other	Heart Disease	NONE	01/15/2020	140/90	180	name@hospital.org	nan		
10	40	Female	None	ALBUTEROL	20/02/20	140/90	220	patient@example.com	555-555-5555		
11	25	Male	Asthma	ATORVASTATIN	April 5, 2018	120/80	200	nan			
12		Male	None	ATORVASTATIN	20/02/20	130/85	200	name@hospital.org			
13	40	Male	Diabetes	ALBUTEROL	03-25-2019	140/90	180	nan	123-456-7890		
14	25	Female	Asthma	ALBUTEROL	03-25-2019	130/85	NaN	name@hospital.org	nan		
15	70	Female	Hypertension	ALBUTEROL	2019.12.01	130/85	220				
16	35	Female	Heart Disease	ATORVASTATIN	03-25-2019	110/70	180		123-456-7890		
17	60	Female	Asthma	ALBUTEROL	20/02/20	120/80	200				
18	70	Female	Asthma	METFORMIN	03-25-2019	130/85	160	contact@domain.com	098-765-4321		
19	25	Other	Diabetes	NONE	01/15/2020	120/80	NaN	name@hospital.org	098-765-4321		
20	70	Male	Heart Disease	NONE	03-25-2019	110/70	160	contact@domain.com			
21	35	Male	Hypertension	LISINAPRIL	April 5, 2018	120/80	180	nan	nan		
22	25	Male	Asthma	LISINAPRIL	01/15/2020	140/90	200	contact@domain.com	098-765-4321		
23	70	Male	Asthma	NONE	2019.12.01	130/85	220	nan	555-555-5555		
24	60	Male	Diabetes	METFORMIN	April 5, 2018	120/80	160		555-555-5555		
25	60	Male	None	ATORVASTATIN	20/02/20	220	220	nan	123-456-7890		
26	40	Male	Hypertension	ATORVASTATIN	20/02/20	120/80	200	contact@domain.com			
27	70	Male	Heart Disease	METFORMIN	20/02/20	130/85	160	name@hospital.org			
28		Other	Asthma	ATORVASTATIN	April 5, 2018	110/70	220	contact@domain.com	098-765-4321		
29	70	Male	Asthma	ALBUTEROL	April 5, 2018	110/70	220	name@hospital.org	nan		
30	25	Other	Hypertension	NONE	April 5, 2018	130/85	160	name@hospital.org			
31	25	Female	None	NONE	01/15/2020	140/90	200	nan	098-765-4321		
32	70	Male	Asthma	METFORMIN	01/15/2020	140/90	220	contact@domain.com	098-765-4321		
33	35	Female	Hypertension	ATORVASTATIN	20/02/20	110/70	180	name@hospital.org			
34	60	Female	Diabetes	LISINAPRIL	03-25-2019	110/70	160		098-765-4321		
35	25	Female	None	LISINAPRIL	20/02/20	130/85	NaN	name@hospital.org	123-456-7890		
36	60	Other	Heart Disease	ATORVASTATIN	2019.12.01	140/90	NaN		123-456-7890		
37	25	Female	Asthma	LISINAPRIL	01/15/2020	110/70	200	contact@domain.com			
38	40	Female	Hypertension	METFORMIN	April 5, 2018	140/90	180	contact@domain.com	nan		

Process: IF()

TRIM	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Blood Pressure	Cholesterol	Email	Phone Number													
1	140/90	200	name@hospital.org	555-555-5555													
2	120/80	200	nan	nan													
3	110/70	160	contact@domain.com	nan													
4	110/70	NaN	name@hospital.org	555-555-5555													
5	110/70	180															
6	140/90	180	nan	nan													
7	120/80	NaN	name@hospital.org	nan													
8		NaN	name@hospital.org	nan													
9	140/90	180	name@hospital.org	nan													
10	140/90	220	patient@example.com	555-555-5555													
11	120/80	200	nan														
12	130/85	200	name@hospital.org														
13	140/90	180	nan	123-456-7890													
14	130/85	NaN	name@hospital.org														
15	130/85	220		nan													
16	110/70	180		123-456-7890													
17	120/80	200															
18	130/85	160	contact@domain.com	098-765-4321													
19	120/80	NaN	name@hospital.org	098-765-4321													
20	110/70	160	contact@domain.com														
21	120/80	180	nan	nan													
22	140/90	200	contact@domain.com	098-765-4321													
23	130/85	220	nan	555-555-5555													
24	120/80	160	nan	555-555-5555													
25		220	nan	123-456-7890													
26	120/80	200	contact@domain.com														
27	130/85	160	name@hospital.org														
28	110/70	220	contact@domain.com	098-765-4321													
29	110/70	220	name@hospital.org	nan													
30	130/85	160	name@hospital.org														
31	140/90	200	nan	098-765-4321													
32	140/90	220	contact@domain.com	098-765-4321													
33	110/70	180	name@hospital.org														
34	110/70	160		098-765-4321													
35	130/85	NaN	name@hospital.org	123-456-7890													
36	140/90	NaN		123-456-7890													
37	110/70	200	contact@domain.com														
38	140/90	180	contact@domain.com	nan													

6) Email & Phone Number

Remove nan and keep blank columns as is

	F	G	H	I	J	K	L	M	N	O	P	Q
1	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number							
2	01/15/2020	140/90	200	name@hospital.org	555-555-5555							
3	April 5, 2018	120/80	200	nan	nan							
4	2019.12.01	110/70	160	contact@domain.com	nan							
5	01/15/2020	110/70		name@hospital.org	555-555-5555							
6	20/02/20	110/70	180									
7	03-25-2019	140/90	180	nan	nan							
8	20/02/20	120/80		name@hospital.org	nan							
9	20/02/20			name@hospital.org	nan							
10	01/15/2020	140/90	180	name@hospital.org	nan							
11	20/02/20	140/90	220	patient@example.com	555-555-5555							
12	April 5, 2018	120/80	200	nan								
13	20/02/20	130/85	200	name@hospital.org								
14	03-25-2019	140/90	180	nan	123-456-7890							
15	03-25-2019	130/85		name@hospital.org	nan							
16	2019.12.01	130/85	220		nan							
17	03-25-2019	110/70	180		123-456-7890							
18	20/02/20	120/80	200									
19	03-25-2019	130/85	160	contact@domain.com	098-765-4321							
20	01/15/2020	120/80		name@hospital.org	098-765-4321							
21	03-25-2019	110/70	160	contact@domain.com								
22	April 5, 2018	120/80	180	nan	nan							
23	01/15/2020	140/90	200	contact@domain.com	098-765-4321							
24	2019.12.01	130/85	220	nan	555-555-5555							
25	April 5, 2018	120/80	160	nan	555-555-5555							
26	20/02/20		220	nan	123-456-7890							
27	20/02/20	120/80	200	contact@domain.com								
28	20/02/20	130/85	160	name@hospital.org								
29	April 5, 2018	110/70	220	contact@domain.com	098-765-4321							
30	April 5, 2018	110/70	220	name@hospital.org	nan							
31	April 5, 2018	130/85	160	name@hospital.org								
32	01/15/2020	140/90	200	nan	098-765-4321							
33	01/15/2020	140/90	220	contact@domain.com	098-765-4321							
34	20/02/20	110/70	180	name@hospital.org								
35	03-25-2019	110/70	160		098-765-4321							
36	20/02/20	130/85		name@hospital.org	123-456-7890							
37	2019.12.01	140/90			123-456-7890							
38	01/15/2020	110/70	200	contact@domain.com								
39	April 5, 2018	140/90	180	contact@domain.com	nan							

Process: IF(), OR()

	E	F	G	H	I	J	K	L	M	N	O
1	Medication	Visit Date	Blood Pressure	Cholesterol	Email	Phone Number					
2	Metformin	01/15/2020	140/90	200	name@hospital.org	555-555-5555		=IF(OR(I2="", LOWER(I2)="nan"), "", I2)			
3	None	April 5, 2018	120/80	200	nan	nan					
4	Metformin	2019.12.01	110/70	160	contact@domain.com	nan		contact@domain.com			
5	Albuterol	01/15/2020	110/70		name@hospital.org	555-555-5555		name@hospital.org			
6	None	20/02/20	110/70	180							
7	Metformin	03-25-2019	140/90	180	nan	nan					
8	Lisinopril	20/02/20	120/80		name@hospital.org	nan		name@hospital.org			
9	None	20/02/20			name@hospital.org	nan		name@hospital.org			
10	None	01/15/2020	140/90	180	name@hospital.org	nan		name@hospital.org			
11	Albuterol	20/02/20	140/90	220	patient@example.com	555-555-5555		patient@example.com			
12	Atorvastatin	April 5, 2018	120/80	200	nan						
13	Atorvastatin	20/02/20	130/85	200	name@hospital.org			name@hospital.org			
14	Albuterol	03-25-2019	140/90	180	nan	123-456-7890					
15	Albuterol	03-25-2019	130/85		name@hospital.org	nan		name@hospital.org			
16	Albuterol	2019.12.01	130/85	220		nan					
17	Atorvastatin	03-25-2019	110/70	180		123-456-7890					
18	Albuterol	20/02/20	120/80	200							
19	Metformin	03-25-2019	130/85	160	contact@domain.com	098-765-4321		contact@domain.com			
20	None	01/15/2020	120/80		name@hospital.org	098-765-4321		name@hospital.org			
21	None	03-25-2019	110/70	160	contact@domain.com			contact@domain.com			
22	Lisinopril	April 5, 2018	120/80	180	nan						
23	Lisinopril	01/15/2020	140/90	200	contact@domain.com	098-765-4321		contact@domain.com			
24	None	2019.12.01	130/85	220	nan	555-555-5555					
25	Metformin	April 5, 2018	120/80	160	nan	555-555-5555					
26	Atorvastatin	20/02/20		220	nan	123-456-7890					
27	Atorvastatin	20/02/20	120/80	200	contact@domain.com			contact@domain.com			
28	Metformin	20/02/20	130/85	160	name@hospital.org			name@hospital.org			
29	Atorvastatin	April 5, 2018	110/70	220	contact@domain.com	098-765-4321		contact@domain.com			
30	Albuterol	April 5, 2018	110/70	220	name@hospital.org	nan		name@hospital.org			
31	None	April 5, 2018	130/85	160				name@hospital.org			
32	None	01/15/2020	140/90	200	nan	098-765-4321					
33	Metformin	01/15/2020	140/90	220	contact@domain.com	098-765-4321		contact@domain.com			
34	Atorvastatin	20/02/20	110/70	180	name@hospital.org			name@hospital.org			
35	Lisinopril	03-25-2019	110/70	160		098-765-4321					
36	Lisinopril	20/02/20	130/85		name@hospital.org	123-456-7890		name@hospital.org			
37	Atorvastatin	2019.12.01	140/90			123-456-7890					
38	Lisinopril	01/15/2020	110/70	200	contact@domain.com			contact@domain.com			
39	Metformin	April 5, 2018	140/90	180	contact@domain.com	nan		contact@domain.com			

7) Visit Date

Keep only '/' formatted string and into mm-dd-yyy format

F1		B	C	D	E	F	G	H	I	J	K
		Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Emails	Phone Number	
1	me	25	Male	Heart Disease	Metformin	01/15/2020	140/90	200	name@hospital.org	555-555-5555	
2	ne	35	Male	Diabetes	None	April 5, 2018	120/80	200	name@hospital.org		
3	ne		Other	Asthma	Metformin	2019.12.01	110/70	160	contact@domain.com		
4	net		Male	Diabetes	Albuterol	01/15/2020	110/70		name@hospital.org	555-555-5555	
5	son		Female	Asthma	None	20/02/20	110/70	180			
6	ne		Male	Hypertension	Metformin	03-25-2019	140/90	180			
7	rk		Male	Hypertension	Lisinopril	20/02/20	120/80		name@hospital.org		
8	wn	60	Other	None	None	20/02/20			name@hospital.org		
9	ne	60	Other	Heart Disease	None	01/15/2020	140/90	180			
10	nez	40	Female	None	Albuterol	20/02/20	140/90	220	patient@example.com	555-555-5555	
11	son	25	Male	Asthma	Atorvastatin	April 5, 2018	120/80	200			
12	ne	25	Male	None	Atorvastatin	20/02/20	130/85	200	name@hospital.org		
13	nez	40	Male	Diabetes	Albuterol	03-25-2019	140/90	180		123-456-7890	
14	wn	25	Female	Asthma	Albuterol	03-25-2019	130/85		name@hospital.org		
15	son	70	Female	Hypertension	Albuterol	2019.12.01	130/85	220			
16	son	35	Female	Heart Disease	Atorvastatin	03-25-2019	110/70	180		123-456-7890	
17	h	60	Female	Asthma	Albuterol	20/02/20	120/80	200			
18	nez	70	Female	Asthma	Metformin	03-25-2019	130/85	160	contact@domain.com	098-765-4321	
19	son	25	Other	Diabetes	None	01/15/2020	120/80		name@hospital.org	098-765-4321	
20	h	70	Male	Heart Disease	None	03-25-2019	110/70	160	contact@domain.com		
21	son	35	Male	Hypertension	Lisinopril	April 5, 2018	120/80	180			
22	ne	25	Male	Asthma	Lisinopril	01/15/2020	140/90	200	contact@domain.com	098-765-4321	
23	rk	70	Male	Asthma	None	2019.12.01	130/85	220		555-555-5555	
24	son	60	Male	Diabetes	Metformin	April 5, 2018	120/80	160		555-555-5555	
25	lor	60	Male	None	Atorvastatin	20/02/20	220			123-456-7890	
26	ne	40	Male	Hypertension	Atorvastatin	20/02/20	120/80	200	contact@domain.com		
27	rk	70	Male	Heart Disease	Metformin	20/02/20	130/85	160	name@hospital.org		
28	wn		Other	Asthma	Atorvastatin	April 5, 2018	110/70	220	contact@domain.com	098-765-4321	
29	ne	70	Male	Asthma	Albuterol	April 5, 2018	110/70	220	name@hospital.org		
30	is	25	Other	Hypertension	None	April 5, 2018	130/85	160	name@hospital.org		
31	lor	25	Female	None	None	01/15/2020	140/90	200		098-765-4321	
32	wn	70	Male	Asthma	Metformin	01/15/2020	140/90	220	contact@domain.com	098-765-4321	
33	son	35	Female	Hypertension	Atorvastatin	20/02/20	110/70	180	name@hospital.org		
34	wn	60	Female	Diabetes	Lisinopril	03-25-2019	110/70	160		098-765-4321	
35	ne	25	Female	None	Lisinopril	20/02/20	130/85		name@hospital.org	123-456-7890	
36	son	60	Other	Heart Disease	Atorvastatin	2019.12.01	140/90			123-456-7890	
37	lor	25	Female	Asthma	Lisinopril	01/15/2020	110/70	200	contact@domain.com		
38	ne	40	Female	Hypertension	Metformin	April 5, 2018	140/90	180	contact@domain.com		

Process:

- 1) Find / formatted dates using SEARCH() & ISNUMBER()
- 2) Convert Text to Column in the Data Tab
- 3) Select a custom format using Cmd + 1

	F	G	H	I	J	K	L	M	N	O	P	Q
1	Visit Date	Blood Pressure	Cholesterol	Emails	Phone Number							
2	01/15/2020	140/90	200	name@hospital.org	555-555-5555		=IF(ISNUMBER(SEARCH("J",F2)),F2,"")					
3	April 5, 2018	120/80	200									
4	2019.12.01	110/70	160	contact@domain.com								
5	01/15/2020	110/70		name@hospital.org	555-555-5555							
6	20/02/20	110/70	180									
7	03-25-2019	140/90	180									
8	20/02/20	120/80		name@hospital.org								
9	20/02/20			name@hospital.org								
10	01/15/2020	140/90	180	name@hospital.org								
11	20/02/20	140/90	220	patient@example.com	555-555-5555							
12	April 5, 2018	120/80	200									
13	20/02/20	130/85	200	name@hospital.org								
14	03-25-2019	140/90	180									
15	03-25-2019	130/85		name@hospital.org	123-456-7890							
16	2019.12.01	130/85	220									
17	03-25-2019	110/70	180		123-456-7890							
18	20/02/20	120/80	200									
19	03-25-2019	130/85	160	contact@domain.com	098-765-4321							
20	01/15/2020	120/80		name@hospital.org	098-765-4321							
21	03-25-2019	110/70	160	contact@domain.com								
22	April 5, 2018	120/80	180									
23	01/15/2020	140/90	200	contact@domain.com	098-765-4321							
24	2019.12.01	130/85	220		555-555-5555							
25	April 5, 2018	120/80	160		555-555-5555							
26	20/02/20		220		123-456-7890							
27	20/02/20	120/80	200	contact@domain.com								
28	20/02/20	130/85	160	name@hospital.org								
29	April 5, 2018	110/70	220	contact@domain.com	098-765-4321							
30	April 5, 2018	110/70	220	name@hospital.org								
31	April 5, 2018	130/85	160	name@hospital.org								
32	01/15/2020	140/90	200		098-765-4321							
33	01/15/2020	140/90	220	contact@domain.com	098-765-4321							
34	20/02/20	110/70	180	name@hospital.org								
35	03-25-2019	110/70	160		098-765-4321							
36	20/02/20	130/85		name@hospital.org	123-456-7890							
37	2019.12.01	140/90			123-456-7890							
38	01/15/2020	110/70	200	contact@domain.com								
39	April 5, 2018	140/90	180	contact@domain.com								

AutoSave healthcare_data_exp1 Search (Cmd + Ctrl + U)

Home Insert Draw Page Layout Formulas Data Review View

Calibri (Body) 12 A A Wrap Text Custom Conditional Formatting Cell Styles Insert Delete Format Sort & Filter Find & Select Analyse Data

F4SS

	A	B	C	D	E	F	G	H	I	J
	Patient Name	Age	Gender	Condition	Medication	Visit Date	Blood Pressure	Cholesterol	Emails	Phone Number
1	David Lee	25	Other	Heart Disease	Metformin	01-15-2020	140/90	200	name@hospital.org	555-555-5555
2	Emily Davis		Male	Diabetes	None		120/80	200		
3	Laura Martinez	35	Other	Asthma	Metformin		110/70	160	contact@domain.com	
4	Michael Wilson		Male	Diabetes	Albuterol	01-15-2020	110/70		name@hospital.org	555-555-5555
5	David Lee		Female	Asthma	None		110/70	180		
6	Mary Clark		Male	Hypertension	Metformin		140/90	180		
7	Robert Brown		Male	Hypertension	Lisinopril		120/80		name@hospital.org	
8	David Lee	60	Other	None	None				name@hospital.org	
9	Laura Martinez	60	Other	Heart Disease	None	01-15-2020	140/90	180	name@hospital.org	
10	Michael Wilson	40	Female	None	Albuterol		140/90	220	patient@example.com	555-555-5555
11	Emily Davis	25	Male	Asthma	Atorvastatin		120/80	200		
12	Laura Martinez		Male	None	Atorvastatin		130/85	200	name@hospital.org	
13	Laura Martinez	40	Male	Diabetes	Albuterol		140/90	180		123-456-7890
14	Robert Brown	25	Female	Asthma	Albuterol		130/85		name@hospital.org	
15	Sarah Johnson	70	Female	Hypertension	Albuterol		130/85	220		
16	Michael Wilson	35	Female	Heart Disease	Atorvastatin		110/70	180		123-456-7890
17	Jane Smith	60	Female	Asthma	Albuterol		120/80	200		
18	Laura Martinez	70	Female	Asthma	Metformin		130/85	160	contact@domain.com	098-765-4321
19	Sarah Johnson	25	Other	Diabetes	None	01-15-2020	120/80		name@hospital.org	098-765-4321
20	Jane Smith	70	Male	Heart Disease	None		110/70	160	contact@domain.com	
21	Michael Wilson	35	Male	Hypertension	Lisinopril		120/80	180		
22	John Doe	25	Male	Asthma	Lisinopril	01-15-2020	140/90	200	contact@domain.com	098-765-4321
23	Mary Clark	70	Male	None	None		130/85	220		555-555-5555
24	Sarah Johnson	60	Male	Diabetes	Metformin		120/80	160		555-555-5555
25	James Taylor	60	Male	None	Atorvastatin			220		123-456-7890
26	John Doe	40	Male	Hypertension	Atorvastatin		120/80	200	contact@domain.com	
27	Mary Clark	70	Male	Heart Disease	Metformin		130/85	160	name@hospital.org	
28	Robert Brown		Other	Asthma	Atorvastatin		110/70	220	contact@domain.com	098-765-4321
29	David Lee	70	Male	Asthma	Albuterol		110/70	220	name@hospital.org	
30	Emily Davis	25	Other	Hypertension	None		130/85	160	name@hospital.org	
31	James Taylor	25	Female	None	None	01-15-2020	140/90	200		098-765-4321
32	Robert Brown	70	Male	Asthma	Metformin	01-15-2020	140/90	220	contact@domain.com	098-765-4321
33	Michael Wilson	35	Female	Hypertension	Atorvastatin		110/70	180	name@hospital.org	
34	Robert Brown	60	Female	Diabetes	Lisinopril		110/70	160		098-765-4321
35	David Lee	25	Female	None	Lisinopril		130/85			123-456-7890
36	Michael Wilson	60	Other	Heart Disease	Atorvastatin		140/90		name@hospital.org	123-456-7890
37	James Taylor	25	Female	Asthma	Lisinopril	01-15-2020	110/70	200	contact@domain.com	
38	David Lee	40	Female	Hypertension	Metformin		140/90	180	contact@domain.com	
39	Jane Smith		Female	Diabetes	Metformin	01-15-2020	110/70	220	contact@domain.com	123-456-7890
40	Emily Davis	25	Male	Heart Disease	Atorvastatin		120/80		name@hospital.org	123-456-7890

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Ready Accessibility: Good to go Average: 01-15-2020 Count: 186 100%

Post Laboratory Questions:

Following questions are to be handwritten and attach the photograph of the answer in this document.

1. Need of Data cleaning in Data Analytics

Data cleaning is a crucial step in the data analytics process because raw data collected from various sources such as surveys, sensors, databases, and web applications is often **incomplete, inconsistent, inaccurate, and noisy**. If this unprocessed data is directly used for analysis, it can lead to **incorrect interpretations, misleading patterns, and faulty decision-making**.

It helps in **identifying and correcting errors**, such as missing values, duplicate records, incorrect entries, and inconsistent formats. It ensures that the dataset is **uniform and standardized**, making it easier to analyze and compare. Clean data improves the **accuracy and reliability** of statistical analysis, machine learning models, and predictive systems.

Moreover, data cleaning enhances **data integrity and quality**, reduces processing time, and increases the overall efficiency of analytical workflows. By ensuring that the data is relevant, complete, and error-free, data cleaning enables organizations and analysts to derive **meaningful insights and informed decisions**, making it an essential foundation for successful data analytics.

2. List any four common data quality issues found in raw datasets.

- **Missing values** – Some data fields may be empty or contain placeholders like NULL or NaN, which can affect analysis results.
- **Duplicate data** – Multiple records representing the same entity can cause biased or misleading outcomes.
- **Inconsistent data formats** – Data stored in different formats, such as dates in multiple styles or mixed units of measurement.
- **Incorrect or invalid data** – Values that are logically incorrect or outside acceptable ranges, such as negative age values or text in numeric columns.